

ALEXANDER DORNBACK

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MS, Robotics @ Columbia | Controls, Manipulation, Sim2Real | Available now for full-time roles

EDUCATION

Columbia University MS, Mechanical Engineering , 3.96/4 Robotics, Automation, and Controls Concentration <i>Research Affiliation: Creative Machines Laboratory, Prof. Hod Lipson (January 2026 – Present)</i>	New York, NY Expected Dec 2026
Case Western Reserve University BS, Biomedical Engineering Biomechanics Concentration Minor, Music	Cleveland, OH May 2022

WORK EXPERIENCE

Serpent Robotics Mechatronics Intern	Philadelphia, PA June 2026 – August 2026
- Owned mechanical design, motor control, ROS 2, and micro-ROS firmware for XY positioning of a cable-driven parallel robot - Designed a 100Hz P-control capstan winch pre-tensioner, unlocking unlimited rope length as a core product differentiator - Architected HALow communication between Base/Robot/Winch Pis and gamepad with watchdogs/state machine interlocks - Diagnosed a descent failure to a BLDC bank bug and shipped a periodic re-assert fix, also resolving a related saw actuation fault - Built a MuJoCo digital twin and kinematic solver with Jacobian singularity analysis as a positioning algorithm for auto-coursing	
Weill Cornell Medicine Dario Institute Robotics Engineer	New York, NY Jul 2022 - Sep 2025
- Led 3 engineers over 12 months to deliver a clinically validated intubation robot, managing deployment testing with ENTs - Designed a superelastic tendon-driven actuator and transmission mechanism, achieving 30°/s tip deflection speeds - Embedded C++ (Arduino) for IR closed-loop control achieved first-pass intubation in 20-second benchmark with 90% accuracy - Created an image segmentation protocol for converting 65 CT angiograms into CFD-validated multi-material 3D models	
Haima Therapeutics Formulation Scientist	Cleveland, OH Oct 2020 - Jun 2022
Characterized 200+ nanoparticle batches for a \$46M DARPA FSHARP grant; co-author, Sekhon et al. Wiley Adv. Science, 2025	

ENGINEERING PROJECTS

FIDO: an open-source, trainable, 3D-printed quadruped robot (Prof. Hod Lipson)
- Implemented random search/hill-climbing for gait parameter optimization and PPO for sensorless joint neutral estimation - Achieved stance in 10 minutes of parallel-environment PPO training on 4 custom linkage-drive legs in MuJoCo - Reached stable walking in 30-minute training and completed sim-to-real pipeline for gait of 25 cm/s on real-world deployment
CATCH: Hybrid MPC-CBF impact shaping for bipedal locomotion (Prof. Homayoon Beigi)
- Selected CBF after PID and vanilla MPC exceeded 3.34N actuator limits on impact; enforced safety across drops up to 40cm - Built kinematic TTC estimator feeding a dynamic MPC horizon, achieving 16% GRF reduction at 40cm within hardware limits
GRASP: Perception-to-action manipulation pipeline (Viam Hackathon / Independent)
- Fine-tuned a customized detection model on 100 hand-labeled images, achieving 95% confidence at 8.3 FPS to drive an xArm 6 - Rebuilt perception in MuJoCo with YOLOv8-seg on 4k domain-randomized renders and PCA-based grasp-axis extraction - Extended a 5-DOF IK solver to 6-DOF (grip axis) with tip-anchored rotation and live slip correction for <6mm tip placement error

SKILLS

- Learning: Reinforcement Learning (PPO), Behavioral Cloning, Diffusion Policy, Domain Randomization
- Controls: Model Predictive Control, Control Barrier Functions, PID, Sim2Real
- Software: Python, ROS 2, C++, MATLAB/Simulink, I2C/SPI, Linux, Git
- Mechatronics: Embedded C++ (ESP32/Arduino/Teensy), Sensor Fusion, Actuator/Motor Control, Transmission Design
- Modeling and Design: MuJoCo, Gazebo, Kinematics (FK/IK), State Estimation, Forward Modeling, SolidWorks (CSWA, Sim)